$$P'(x) = 2\left(1-\frac{k}{x^2}\right)$$

$$P'(x) = 0 \Rightarrow x = \pm\sqrt{k}\,.$$

As x, y are sides of the rectangle, $x = \sqrt{k}$ is a critical number.

Now, $P''(x) = \frac{4k}{x^3}$ and $P''(\sqrt{k}) > 0 \Rightarrow P(x)$ has its minimum value at $x = \sqrt{k}$.

Substituting $x = \sqrt{k}$ in $xy = k$ we get $y = \sqrt{k}$. Therefore the minimum perimeter rectangle of a given area is a square. ■

EXERCISE 7.8

1. Find two positive numbers whose sum is 12 and their product is maximum.
2. Find two positive numbers whose product is 20 and their sum is minimum.
3. Find the smallest possible value of $x^2 + y^2$ given that $x + y = 10$.
4. A garden is to be laid out in a rectangular area and protected by wire fence. What is the largest possible area of the fenced garden with 40 metres of wire.
5. A rectangular page is to contain 24 cm^2 of print. The margins at the top and bottom of the page are 1.5 cm and the margins at other sides of the page is 1 cm. What should be the dimensions of the page so that the area of the paper used is minimum.
6. A farmer plans to fence a rectangular pasture adjacent to a river. The pasture must contain 1,80,000 sq.mtrs in order to provide enough grass for herds. No fencing is needed along the river. What is the length of the minimum needed fencing material?
7. Find the dimensions of the rectangle with maximum area that can be inscribed in a circle of radius 10 cm.
8. Prove that among all the rectangles of the given perimeter, the square has the maximum area.
9. Find the dimensions of the largest rectangle that can be inscribed in a semi circle of radius r cm.
10. A manufacturer wants to design an open box having a square base and a surface area of 108 sq.cm. Determine the dimensions of the box for the maximum volume.
11. The volume of a cylinder is given by the formula $V = \pi r^2 h$. Find the greatest and least values of V if $r + h = 6$.
12. A hollow cone with base radius a cm and height b cm is placed on a table. Show that the volume of the largest cylinder that can be hidden underneath is $\frac{4}{9}$ times volume of the cone.

7.9 Symmetry and Asymptotes

7.9.1 Symmetry

Consider the following curves and observe that each of them is having some special properties, called symmetry with respect to a point, with respect to a line.

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